

Student Marks Analysis — Project Report

STUDENT MARKS ANALYSIS

Python Data Processing & Basic Statistical Analysis

Task 1 — AI/ML Track

Submitted by:

Antara Sarvade

Technology Used:

Python, Jupyter Notebook

Project Type:

Mini Project / Internship Task

Date:

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Introduction

Student Marks Analysis is a beginner-level Python data processing project developed to analyze the marks obtained by a group of students.

The project demonstrates how Python's built-in functions can be used to perform basic statistical calculations on a dataset. The analysis focuses on determining the number of students, total marks, average marks, highest marks, and lowest marks.

This project provides a foundation for understanding basic data processing concepts before moving toward more advanced data analysis using libraries such as **NumPy and Pandas**.

Problem Statement

The objective of this project is to analyze a dataset containing marks obtained by students and calculate important statistical measures.

The analysis should determine:

- Number of students in the dataset
- Total marks obtained
- Average marks
- Highest marks
- Lowest marks

The results should be displayed in a clear and understandable format.

Objectives

The main objectives of this project are:

1. To store student marks using a Python list.
2. To determine the total number of students.
3. To calculate the total marks obtained.
4. To calculate the average marks.
5. To identify the highest marks.
6. To identify the lowest marks.
7. To practice Python's built-in functions for basic data processing.
8. To understand the foundation of statistical analysis using Python.

Dataset

For this project, a manually created dataset containing marks of **10 students** was used.

Dataset

Student	Marks
Student 1	78
Student 2	85
Student 3	92
Student 4	67
Student 5	88
Student 6	74
Student 7	95
Student 8	81
Student 9	69
Student 10	90

The marks were stored in Python using a list:

```
marks = [7]
```

Methodology

The project follows a simple data processing workflow.

Step 1 — Store the Dataset

The student marks are stored in a Python list.

Step 2 — Count Students

The `len()` function is used to determine the number of students.

```
number_of_students = len(marks)
```

Step 3 — Calculate Total Marks

The `sum()` function is used to calculate the combined marks.

```
total_marks = sum(marks)
```

Step 4 — Calculate Average

The average is calculated using the total marks divided by the number of students.

Formula:

$$\text{Average Marks} = \text{Total Marks} / \text{Number of Students}$$

```
average_marks = total_marks / number_of_students
```

Step 5 — Find Highest Marks

The `max()` function identifies the highest mark.

```
highest_marks = max(marks)
```

Step 6 — Find Lowest Marks

The `min()` function identifies the lowest mark.

```
lowest_marks = min(marks)
```

Tools and Technologies Used

Technology / Tool	Purpose
Python	Data processing and calculations
Jupyter Notebook	Writing and executing Python code
Python Lists	Storing student marks
Built-in Functions	Performing statistical calculations

Python Functions Used

Function	Purpose
<code>len()</code>	Determines the number of elements
<code>sum()</code>	Calculates the total

<code>max()</code>	Finds the highest value
<code>min()</code>	Finds the lowest value

Implementation

The complete Python implementation combines all the calculations into a single analysis.

```
# Student Marks Analysis
```

```
marks = [78, 85, 92, 67, 88, 74, 95, 81, 69, 90]
```

```
# Basic calculations
```

```
number_of_students = len(marks)
```

```
total_marks = sum(marks)
```

```
average_marks = total_marks / number_of_students
```

```
highest_marks = max(marks)
```

```
lowest_marks = min(marks)
```

```
# Display results
```

```
print("=" * 40)
```

```
print("    STUDENT MARKS ANALYSIS")
```

```
print("=" * 40)
```

```
print(f'Number of Students : {number_of_students}')
```

```
print(f'Total Marks      : {total_marks}')
```

```
print(f'Average Marks    : {average_marks:.2f}')
```

```
print(f'Highest Marks    : {highest_marks}')
```

```
print(f'Lowest Marks     : {lowest_marks}')
```

```
print("=" * 40)
```

Results

The analysis produced the following results:

Metric	Result
Number of Students	10
Total Marks	819
Average Marks	81.90
Highest Marks	95
Lowest Marks	67

Output

```
=====
```

STUDENT MARKS ANALYSIS

Number of Students : 10
Total Marks : 819
Average Marks : 81.90
Highest Marks : 95
Lowest Marks : 67

Analysis and Observations

Based on the calculated results, the following observations were made:

- The dataset contains marks for **10 students**.
- The combined marks obtained by all students are **819**.
- The average mark is **81.90**.
- The highest mark recorded is **95**.
- The lowest mark recorded is **67**.
- The difference between the highest and lowest marks is **28 marks**.

The results demonstrate how basic Python operations can be used to quickly summarize numerical data.

Learning Outcomes

Through this project, the following concepts were practiced:

- Creating and working with Python lists
- Using built-in Python functions
- Performing arithmetic calculations
- Working with numerical datasets
- Calculating basic statistical measures
- Formatting program output
- Organizing Python code into a structured analysis

The project also provided an introduction to concepts that are commonly used in data analysis and machine learning workflows.

Future Scope

This project can be extended in several ways to perform more advanced analysis.

Possible improvements include:

- Using **NumPy** for numerical operations.
- Using **Pandas DataFrames** to manage student data.
- Adding student names and subjects.
- Calculating subject-wise averages.
- Assigning grades based on marks.
- Visualizing marks using charts and graphs.
- Reading student data from CSV or Excel files.
- Performing more detailed statistical analysis.

The next stage of development can involve converting this Python-list-based analysis into a **Pandas-based data analysis project**.

The Student Marks Analysis project successfully demonstrates the use of Python for basic data processing and statistical calculations.

Using a simple dataset of 10 students, important measures such as the number of students, total marks, average marks, highest marks, and lowest marks were calculated using Python's built-in functions.

This project helped strengthen the understanding of fundamental Python concepts and provides a foundation for progressing toward more advanced data analysis using **NumPy, Pandas, and data visualization libraries**.

References

You can keep this section very simple:

- Python Documentation — Built-in Functions
- Jupyter Notebook Documentation
- Personal implementation and experimentation